MATH 128 - HW12 - Jie Wen

Problem 1: Numerical Quadrature 1

Seeking a quadrature formula of the form

$$\int_1^3 f(x)dx \approx Af(0) + Bf(1) + Cf(4),$$

which should be **exact for all polynomials of degree ≤ 2**.

Exactness is enforced for $f(x) = 1, x, x^2$.

Let $f(x) = 1$: $\int_1^3 1dx = 2$.
With $f(0) = 1, f(1) = 1, f(4) = 1$:

$$A \cdot 1 + B \cdot 1 + C \cdot 1 = A + B + C = 2. \tag{1}$$

Let $f(x) = x$: $\int_1^3 xdx = \frac{x^2}{2}\Big|_1^3 = \frac{9-1}{2} = 4$.
With $f(0) = 0, f(1) = 1, f(4) = 4$:

$$A \cdot 0 + B \cdot 1 + C \cdot 4 = B + 4C = 4. \tag{2}$$

Let $f(x) = x^2$: $\int_1^3 x^2dx = \frac{x^3}{3}\Big|_1^3 = \frac{27-1}{3} = \frac{26}{3}$.
With $f(0) = 0, f(1) = 1, f(4) = 16$:

$$A \cdot 0 + B \cdot 1 + C \cdot 16 = B + 16C = \frac{26}{3}. \tag{3}$$

Solve for A, B, C:

$$C = \frac{7}{18}.$$

$$B = \frac{22}{9}.$$

$$A = -\frac{5}{6}.$$

$$\therefore \int_1^3 f(x)dx \approx -\frac{5}{6}f(0) + \frac{22}{9}f(1) + \frac{7}{18}f(4)$$

Problem 2: Numerical Quadrature 2

$$L_0(x) = \frac{(x+1)(x-0)}{(-2+1)(-2-0)} = \frac{(x+1)x}{2},$$

$$L_1(x) = \frac{(x+2)(x-0)}{(-1+2)(-1-0)} = -(x+2)x,$$

$$L_2(x) = \frac{(x+2)(x+1)}{(0+2)(0+1)} = \frac{(x+2)(x+1)}{2}.$$

$$P_2(x) = f(-2)L_0(x) + f(-1)L_1(x) + f(0)L_2(x).$$

The Newton–Cotes rule is obtained by integrating $P_2(x)$ from 0 to 1:

$$\int_0^1 f(x)dx \approx \int_0^1 P_2(x)dx = a_0 f(-2) + a_1 f(-1) + a_2 f(0),$$

where

$$a_i = \int_0^1 L_i(x)dx.$$

For a_0:

$$L_0(x) = \frac{(x+1)x}{2} = \frac{x^2+x}{2}.$$

$$a_0 = \int_0^1 L_0(x)dx = \frac{1}{2}\int_0^1 (x^2+x)dx = \frac{1}{2}\left[\frac{x^3}{3}+\frac{x^2}{2}\right]_0^1 = \frac{1}{2}\left(\frac{1}{3}+\frac{1}{2}\right) = \frac{5}{12}.$$

For a_1:

$$L_1(x) = -(x+2)x = -(x^2+2x).$$

$$a_1 = \int_0^1 L_1(x)dx = -\int_0^1 (x^2+2x)dx = -\left[\frac{x^3}{3}+x^2\right]_0^1 = -\left(\frac{1}{3}+1\right) = -\frac{4}{3}.$$

For a_2:

$$L_2(x) = \frac{(x+2)(x+1)}{2} = \frac{x^2+3x+2}{2}.$$

$$a_2 = \int_0^1 L_2(x)dx = \frac{1}{2}\int_0^1 (x^2+3x+2)dx = \frac{1}{2}\left[\frac{x^3}{3}+\frac{3x^2}{2}+2x\right]_0^1 = \frac{1}{2}\left(\frac{1}{3}+\frac{3}{2}+2\right) = \frac{23}{12}.$$

So the **Newton–Cotes formula** based on nodes $-2, -1, 0$ for the integral over $[0,1]$ is

$$\therefore \int_0^1 f(x)dx \approx \frac{5}{12}f(-2) - \frac{4}{3}f(-1) + \frac{23}{12}f(0). \frac{5}{12}(-\sin 2) - \frac{4}{3}(-\sin 1) + \frac{23}{12}\cdot 0 = -\frac{5}{12}\sin 2 + \frac{4}{3}\sin 1$$

Problem 3: Numerical Quadrature 3

Because the rule is linear in f, it suffices that it be exact for each basis function of this 2-dimensional space:

1. $f_1(x) = e^x$
2. $f_2(x) = \cos\left(\frac{\pi x}{2}\right)$

1. Exactness for $f_1(x) = e^x$

True integral:

$$I_{[0,1]}(e^x) = \int_0^1 e^x dx = e - 1.$$

Quadrature approximation:

$$\hat{I}_{[0,1]}(e^x) = w_0 e^0 + w_1 e^1 = w_0 + w_1 e.$$

Exactness condition:

$$w_0 + w_1 e = e - 1. \tag{1}$$

2. Exactness for $f_2(x) = \cos\left(\frac{\pi x}{2}\right)$

True integral:

$$\int_0^1 \cos\left(\frac{\pi x}{2}\right)dx = \frac{2}{\pi}\sin\left(\frac{\pi x}{2}\right)\Big|_0^1 = \frac{2}{\pi}(1-0) = \frac{2}{\pi}.$$

Quadrature approximation:

$$\hat{I}_{[0,1]}\left(\cos\left(\frac{\pi x}{2}\right)\right) = w_0\cos(0) + w_1\cos\left(\frac{\pi}{2}\right) = w_0\cdot 1 + w_1\cdot 0 = w_0.$$

Exactness condition:

$$w_0 = \frac{2}{\pi}. \tag{2}$$

3. Solve for w_1

Substitute $w_0 = \dfrac{2}{\pi}$ into equation (1):

$$w_1 = \frac{e - 1 - \frac{2}{\pi}}{e}.$$

So the desired quadrature rule is

$$\hat{I}_{[0,1]}(f) = \frac{2}{\pi}f(0) + \frac{e - 1 - \frac{2}{\pi}}{e}f(1)$$

Problem 4: Numerical Quadrature 4

a) Quadrature rule on $[0,1]$

f is approximated by the Lagrange interpolation polynomial of degree 1 using the nodes $x_0 = \frac{1}{3}, x_1 = \frac{2}{3}$:

$$P_1(x) = f\left(\tfrac{1}{3}\right)L_0(x) + f\left(\tfrac{2}{3}\right)L_1(x),$$

where

$$L_0(x) = \frac{x - x_1}{x_0 - x_1} = -3x + 2, \qquad L_1(x) = \frac{x - x_0}{x_1 - x_0} = 3x - 1.$$

Using $\displaystyle\int_0^1 f(x)dx \approx \int_0^1 P_1(x)dx$:

$$w_0 = \int_0^1 L_0(x)dx = \int_0^1 (-3x+2)dx = \left[-\frac{3}{2}x^2 + 2x\right]_0^1 = -\frac{3}{2} + 2 = \frac{1}{2},$$

$$w_1 = \int_0^1 L_1(x)dx = \int_0^1 (3x-1)dx = \left[\frac{3}{2}x^2 - x\right]_0^1 = \frac{3}{2} - 1 = \frac{1}{2}.$$

The quadrature rule is

$$\int_0^1 f(x)dx \approx \hat{I}_{[0,1]}(f) = \frac{1}{2}f\left(\tfrac{1}{3}\right) + \frac{1}{2}f\left(\tfrac{2}{3}\right).$$

b) Transformation to a general interval $[a,b]$

Let $x \in [a,b]$ and set the change of variables

$$u = \frac{x-a}{b-a}, \quad \text{so } x = a + (b-a)u, \quad dx = (b-a)du.$$

Then

$$\int_a^b f(x)dx = (b-a)\int_0^1 f\Big[a + (b-a)u\Big]du.$$

Applying the rule from (a) to the function $g(u) = f(a + (b-a)u)$ on $[0,1]$:

$$\int_0^1 g(u)du \approx \frac{1}{2}g\left(\tfrac{1}{3}\right) + \frac{1}{2}g\left(\tfrac{2}{3}\right).$$

Thus

$$\int_a^b f(x)dx \approx (b-a)\cdot\frac{1}{2}\left[f\Big(a+(b-a)\tfrac{1}{3}\Big) + f\Big(a+(b-a)\tfrac{2}{3}\Big)\right] \int_a^b f(x)dx = \frac{b-a}{2}\left[f\left(a+\frac{b-a}{3}\right) + f\left(a+\frac{2(b-a)}{3}\right)\right].$$

Nodes: $x_1 = a + \frac{b-a}{3},\ x_2 = a + \frac{2(b-a)}{3}$; both weights: $\frac{b-a}{2}$.

c) Application to $\int_0^\pi \sin x dx$ and comparison

Here $a = 0, b = \pi$. The formula from (b) yields:

- Nodes:

$$x_1 = 0 + \frac{\pi - 0}{3} = \frac{\pi}{3}, \quad x_2 = 0 + \frac{2(\pi - 0)}{3} = \frac{2\pi}{3}.$$

- Weights: $\dfrac{b-a}{2} = \dfrac{\pi}{2}$.

Approximation:

$$\int_0^\pi \sin x dx \approx \frac{\pi}{2}\left[\sin\left(\frac{\pi}{3}\right) + \sin\left(\frac{2\pi}{3}\right)\right] = \frac{\pi}{2}\left(\frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2}\right) = \frac{\pi\sqrt{3}}{2}.$$

The exact value is

$$\int_0^\pi \sin x dx = [-\cos x]_0^\pi = -\cos\pi + \cos 0 = 1 + 1 = 2.$$

Comparison:

- **Approximate (quadrature):** $\dfrac{\pi\sqrt{3}}{2} \approx 2.72$
- **Exact:** 2

This two-point open rule **overestimates** the integral by about 0.72.